

Complex Numbers

Rectangular vs. Polar: $z = x + iy = |z|e^{i\theta}$

$$|z| = \sqrt{z z^*} = \sqrt{x^2 + y^2} \quad \theta = \arctan\left(\frac{y}{x}\right)$$

$$x = |z| \cos \theta \quad y = |z| \sin \theta$$

Euler's Identity: $e^{i\theta} = \cos \theta + i \sin \theta$

$$\sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i} \quad \cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$$

Complex conjugate: $z^* = x - iy = |z|e^{-i\theta}$

$$(z+w)^* = z^* + w^* \quad (z-w)^* = z^* - w^*$$

$$(zw)^* = z^* w^* \quad \left(\frac{z}{w}\right)^* = \frac{z^*}{w^*}$$

$$z^* = z \text{ means } z \text{ is real}$$

$$z^* = -z \text{ means } z \text{ is complex}$$

$$(z^*)^* = (z^*)^*$$

Vector Spaces

A vector space is a nonempty set $\mathcal{V}$ of objects (elements, vectors) that satisfies the following ten axioms:

Closure Axioms (A1-A2)

A1. Closure under vector addition

For every $x, y \in \mathcal{V}$ there's a unique element (vector) $z \in \mathcal{V}$ — called "the sum of x and y " — denoted by $z = x + y$.

A2. Closure under scalar multiplication

$\forall x \in \mathcal{V}$ and $\forall \alpha \in \mathbb{R}$ (or $\mathbb{C}$) there's a unique $z \in \mathcal{V}$ — called "the product of α and x " — denoted by αx .

Axioms of Vector Addition (A3-A5)

A3. Commutativity of Vector Addition

$\forall x, y \in \mathcal{V}$ we have $x + y = y + x$

A4. Associativity of Vector Addition

$\forall x, y, z \in \mathcal{V}$ $x + (y + z) = (x + y) + z$

A5. Existence of a Zero Element 0

there exists an element in $\mathcal{V}$ — denoted by 0 — such that for every $x \in \mathcal{V}$, $x + 0 = x$

Q: $\mathcal{V} = \{f: \mathbb{R} \rightarrow \mathbb{R} \mid f(t) = _, \forall t \in \mathbb{R}\}$

the zero element in this $\mathcal{V}$ is $f: \mathbb{R} \rightarrow \mathbb{R} \quad \forall t \in \mathbb{R}, f(t) = 0$

* the zero element is unique!!

A6. Existence of Negative

$\forall x \in \mathcal{V}$ the element $(-1)x$ (aka " $-x$ ") has the property that $x + (-1)x = x - x = 0$

* Each element $x \in \mathcal{V}$ has a unique negative.

Axioms for Scalar Multiplication (A7-A10)

A7. Associativity of Scalar Multiplication

$\forall x \in \mathcal{V} \quad \forall \alpha, \beta \in \mathbb{R}$

$$\alpha(\beta x) = (\alpha\beta)x$$

A8. Distributivity of Scalar Multiplication over Vector Addition

$\forall x, y \in \mathcal{V}$ and $\forall \alpha \in \mathbb{R}$

$$\alpha(x + y) = \alpha x + \alpha y$$

A9. Distributivity over Scalar Addition

$\forall x \in \mathcal{V}$ and $\forall \alpha, \beta \in \mathbb{R}$

$$(\alpha + \beta)x = \alpha x + \beta x$$

A10. Existence of Identity

$\forall x \in \mathcal{V}$ we have $1x = x$

scalar vector element in $\mathcal{V}$

Subspace: A subset of a vector space $\mathcal{V}$ ($S \subseteq \mathcal{V}$) that is itself a vector space (in its own right) is called a subspace

* We need to check only 3 things:

a) is 0 in S ?

b) is S closed under vector addition?

c) is S closed under scalar multiplication?

Alternatively:

a) is S nonempty?

b) same as left

c) same as left

Inner Product in $\mathbb{R}^n$

$$\langle \vec{x}, \vec{y} \rangle = \vec{x}^T \vec{y} = x_1 y_1 + \dots + x_n y_n$$

$$\langle \vec{x}, \vec{y} \rangle = \langle \vec{y}, \vec{x} \rangle$$

$$\langle \vec{x}, \vec{x} \rangle = \|\vec{x}\|^2 \geq 0 \text{ w/ equality iff } x=0$$

$$\langle -\vec{x}, \vec{y} \rangle = -\langle \vec{x}, \vec{y} \rangle \text{ or } \langle \alpha \vec{x}, \vec{y} \rangle = \alpha \langle \vec{x}, \vec{y} \rangle$$

$$\langle \vec{x}, \vec{y} + \vec{z} \rangle = \langle \vec{x}, \vec{y} \rangle + \langle \vec{x}, \vec{z} \rangle$$

$$\langle \vec{x} + \vec{y}, \vec{x} + \vec{y} \rangle = \|\vec{x} + \vec{y}\|^2 = \langle \vec{x}, \vec{x} \rangle + 2\langle \vec{x}, \vec{y} \rangle + \langle \vec{y}, \vec{y} \rangle$$

Inner Product in $\mathbb{C}$

$$\langle \vec{x}, \vec{y} \rangle = x_1 y_1^* + \dots + x_n y_n^* = \vec{x}^T \vec{y}^*$$

$$\langle \vec{x}, \vec{y} \rangle = \langle \vec{y}, \vec{x} \rangle^*$$

$$\langle \vec{x}, \vec{x} \rangle = \langle \vec{x}, \vec{x} \rangle^* \Rightarrow \langle x, x \rangle \in \mathbb{R}$$

$$\langle \vec{x}, \vec{y} + \vec{z} \rangle = \langle \vec{x}, \vec{y} \rangle + \langle \vec{x}, \vec{z} \rangle$$

$$\langle \alpha \vec{x}, \vec{y} \rangle = \alpha \langle \vec{x}, \vec{y} \rangle$$

$$\langle x, \beta y \rangle = \beta^* \langle x, y \rangle \quad \forall \beta \in \mathbb{C}$$

Examples:

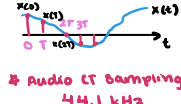
Feature Vector

$$\underline{x} = \begin{bmatrix} 1.72 \text{ m} \\ 76 \text{ kg} \\ 25 \\ 130 \end{bmatrix} \begin{array}{l} \leftarrow \text{height} \\ \leftarrow \text{weight} \\ \leftarrow \text{age} \\ \leftarrow \text{systemic pressure} \end{array}$$

Image Vector

$$\underline{c} = \begin{bmatrix} r \\ g \\ b \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \text{ red } \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \text{ purple}$$

Audio



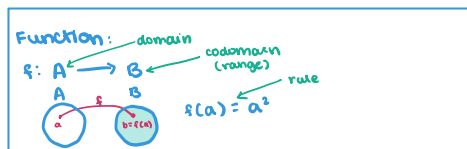
Portfolio Vector

$$\underline{y} = \begin{bmatrix} p_s \\ p_b \\ p_c \end{bmatrix} \begin{array}{l} \leftarrow \text{stocks} \\ \leftarrow \text{bonds} \\ \leftarrow \text{cash} \end{array}$$

$$T = \frac{1}{44100} \text{ sec}$$

* Audio CT sampling

$$\underline{x} = \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_n \end{bmatrix} \begin{array}{l} \leftarrow x(0) \\ \leftarrow x(1) \\ \vdots \\ \leftarrow x(n-1) \end{array}$$



Norm: A function $\|\cdot\|: V \rightarrow \mathbb{R}$ is a norm if for all $x, y \in V$ and all $\alpha \in \mathbb{R}$ (or $\alpha \in \mathbb{C}$):

Triangle Inequality



- $\|x\| \geq 0$ w/ equality iff $x = 0$. *non-negativity*

- $\|\alpha x\| = |\alpha| \cdot \|x\|$ *scaling (homogeneity, non-negative homogeneity)*

- $\|x+y\| \leq \|x\| + \|y\|$ *triangle inequality*

l_1 : homogeneity $\|B\vec{x}\|_1 = |Bx_1| + |Bx_2| + \dots + |Bx_n|$

$$= |B||x_1| + |B||x_2| + \dots + |B||x_n|$$

$$= |B|(|x_1| + |x_2| + \dots + |x_n|)$$

$$= |B| \|\vec{x}\|_1$$

triangle inequality $\|\vec{x} + \vec{y}\|_1 = |x_1 + y_1| + |x_2 + y_2| + \dots + |x_n + y_n|$

$$\leq |x_1| + |y_1| + \dots + |x_n| + |y_n|$$

$$\leq \|\vec{x}\|_1 + \|\vec{y}\|_1$$

nonnegativity each term in $\|\vec{x}\|_1 = |x_1| + \dots + |x_n|$ is

nonnegative and $\|\vec{x}\|_1 = 0$ iff

$$|x_1| = \dots = |x_n| = 0 \text{ which means } \vec{x} = \vec{0}$$

Different Types of Norm

1-norm (l_1 -Norm)

Model the distance a taxicab travels in

a network of perpendicular roadways.

$$\vec{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \quad \|\vec{x}\|_1 = |x_1| + \dots + |x_n| = \sum_{k=1}^n |x_k|$$

∞ -norm (l_∞ -norm)

$\|\vec{x}\|_\infty = \max\{|x_1|, \dots, |x_n|\}$ max distance from a fire station or hospital

2-norm (l_2 -norm)

$\|\vec{x}\|_2 = \sqrt{|x_1|^2 + \dots + |x_n|^2} = \sqrt{\sum_{k=1}^n |x_k|^2}$ magnifies large components; deemphasizes small components

$$\|\vec{x}\|_2^2 = |x_1|^2 + \dots + |x_n|^2 = \sum_{k=1}^n |x_k|^2$$

l_∞ : homogeneity $\|B\vec{x}\|_\infty = \max\{|Bx_1|, \dots, |Bx_n|\}$

$$= \max\{|B||x_1|, \dots, |B||x_n|\}$$

$$= |B| \max\{|x_1|, \dots, |x_n|\}$$

$$= |B| \|\vec{x}\|_\infty$$

triangle inequality $\|\vec{x} + \vec{y}\|_\infty = \max\{|x_1 + y_1|, \dots, |x_n + y_n|\}$

$$\leq \max\{|x_1| + |y_1|, \dots, |x_n| + |y_n|\}$$

$$\leq \max\{|x_1|, \dots, |x_n|\} + \max\{|y_1|, \dots, |y_n|\}$$

$$\leq \|\vec{x}\|_\infty + \|\vec{y}\|_\infty$$

nonnegativity $\|\vec{x}\|_\infty$ is the largest of n nonnegative

numbers $|x_k|$. $\|\vec{x}\|_\infty = 0$ iff $|x_k| = 0$ for

all $k = 1, \dots, n$ which means $\vec{x} = \vec{0}$

Linear Independence

$$\alpha_1 \vec{v}_1 + \dots + \alpha_n \vec{v}_n = \vec{0} \Rightarrow \alpha_1 = \dots = \alpha_n = 0$$

Linear Dependence

$$\alpha_1 \vec{v}_1 + \dots + \alpha_n \vec{v}_n = \vec{0} \text{ for some } \alpha_i \neq 0$$

OR EQUIVALENTLY, $\vec{v}_i = \sum_{j \neq i} \alpha_j \vec{v}_j$ for some $\vec{v}_i$ (one vector is linear combination of others) ** is this vector adding any new information*

STEPS: (shouldn't require systems of equations / 4E)

1. check orthogonality b/t each vector pair

$\hookrightarrow$ if all 0 $\Rightarrow$ linear independent

2. check if any vector $\vec{v}_i$ can be written as a scalar multiple of $\vec{v}_j$

Miscellaneous

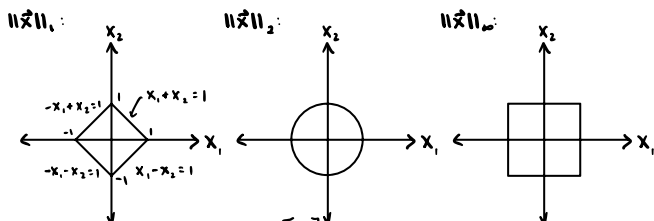
$$\text{dist}(x, y) = \|x - y\| = \|y - x\|$$

$$\text{Define } \tilde{x} = \frac{x}{\|x\|} \quad \tilde{y} = \frac{y}{\|y\|}$$

$$\|\tilde{x}\|^2 = \langle \tilde{x}, \tilde{x} \rangle = \langle \frac{x}{\|x\|}, \frac{x}{\|x\|} \rangle$$

$$= \frac{1}{\|x\|^2} \langle x, x \rangle = \frac{x_1^2 + \dots + x_n^2}{\|x\|^2} = \frac{\|x\|^2}{\|x\|^2} = 1$$

unit ball:



$$\text{Ex: } \langle \mathbf{1}, \vec{x} \rangle = \mathbf{1}^T \vec{x} = [1 \dots 1] \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = x_1 + \dots + x_n = \sum_{k=1}^n x_k$$

$$\text{Ex: } \langle \frac{1}{n} \mathbf{1}, \vec{x} \rangle = \underbrace{[\frac{1}{n} \dots \frac{1}{n}]}_{n\text{-entries}} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \frac{x_1 + \dots + x_n}{n} = \frac{1}{n} \langle \mathbf{1}, \vec{x} \rangle = \frac{1}{n} \sum_{k=1}^n x_k$$

$$\mu_x = \text{avg}(x) = \frac{1}{n} \sum_{k=1}^n x_k \text{ mean of the vector } x$$

$$\text{Ex: } \langle e_k, \vec{x} \rangle = e_k^T \vec{x} = x_k$$

$$e_k = \begin{bmatrix} 0 \\ \vdots \\ 1 \end{bmatrix} \in \mathbb{R}^n$$

$$\{z \in \mathbb{C} \mid |z - 2i| < 3\}$$

$$|z - (a + bi) - 2i| < 3$$

$$|z - (a + (b-2)i)| < 3$$

$$|a + (b-2)i| = 1$$

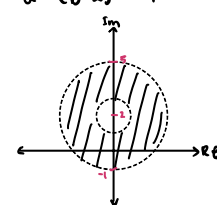
$$\sqrt{a^2 + (b-2)^2} = 1$$

$$a^2 + (b-2)^2 = 1^2$$

$$|a + (b-2)i| = 3$$

$$\sqrt{a^2 + (b-2)^2} = 3$$

$$a^2 + (b-2)^2 = 3^2$$



$$\alpha_1 = \frac{\langle \vec{x}, \vec{v}_1 \rangle}{\langle \vec{v}_1, \vec{v}_1 \rangle}$$

$$\alpha_i = \frac{\langle \vec{x}, \vec{v}_i \rangle}{\langle \vec{v}_i, \vec{v}_i \rangle}$$

$$\cos \theta = \frac{\langle x, y \rangle}{\|x\| \|y\|} = \left\langle \frac{x}{\|x\|}, \frac{y}{\|y\|} \right\rangle$$

Cauchy-Schwarz Inequality

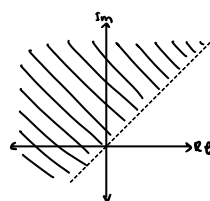
$$|\langle x, y \rangle| \leq \|x\| \|y\|$$

$$-\|x\| \|y\| \leq \langle x, y \rangle \leq \|x\| \|y\|$$

$$-1 \leq \underbrace{\frac{\langle x, y \rangle}{\|x\| \|y\|}}_{\cos \theta_{xy}} \leq 1$$

$\cos \theta_{xy} \triangleq \frac{\langle x, y \rangle}{\|x\| \|y\|} \triangleq$ definition

$$\{z \in \mathbb{C} \mid \text{Im}(z) > \text{Re}(z)\}$$



Eigenvalues & Eigenvectors

$$\vec{v} \rightarrow \boxed{A} \xrightarrow{\lambda \vec{v}} A\vec{v} = \lambda \vec{v} \quad \deg n \quad \# \text{ roots} = n$$

$$\det(\lambda I - A) = 0 \quad \text{"characteristic equation"}$$

Matrix Diagonalization

double change of basis, decoupling system

$$\vec{x} \rightarrow \boxed{A = V\Lambda V^{-1}} \xrightarrow{A\vec{x}} V: \text{coordinates of new basis} \quad n \text{ lin. indep. eigenvectors}$$

$$\text{matrix powers: } A^k = V\Lambda^k V^{-1} \quad \Lambda = V^{-1}AV$$

$$\Lambda^k = \begin{bmatrix} \lambda_1^k & 0 \\ 0 & \lambda_n^k \end{bmatrix}$$

$z = V^{-1}x$ converts to V basis
 $x = Vz$ converts back

$$\text{complex powers: } z = x + iy \Rightarrow z = re^{i\theta} \Rightarrow z^k = r^k e^{ik\theta} \quad r > 1: \text{unstable} \quad r < 1: \text{stable}$$

Projections / Least Squares * least norm (underconstrained): many solutions, but need to choose least norm solution

key equations

$$A\vec{x} = \vec{b} \text{ has no solution}$$

$$A^T A \vec{x} = A^T \vec{b} \quad \text{"normal equation"}$$

$$\vec{\hat{x}} = (A^T A)^{-1} A^T \vec{b} \quad \text{"least squares solution"}$$

$$\vec{p} = A\vec{\hat{x}} = A(A^T A)^{-1} A^T \vec{b} = \text{proj}_A(\vec{b})$$

$$P = A(A^T A)^{-1} A^T \quad \text{"projection matrix"}$$

project onto $C(A)$

$$P = A(A^T A)^{-1} A^T$$

$$p = Pb$$

$$\hat{x} = (A^T A)^{-1} A^T b$$

project onto $C(Q)$

$$P = QQ^T$$

$$p = QQ^T b$$

$$\hat{x} = Q^T b$$

Properties of proj matrices

$$1) P^T = P \quad (P \text{ is "symmetric"})$$

$$P = A(A^T A)^{-1} A^T$$

$$P^T = A((A^T A)^T)^{-1} A^T$$

$$= A((A^T A)^{-1})^T A^T$$

$$= A(A^T A)^{-1} A^T = P$$

orthonormal matrix

orthonormal vector

$$2) P^2 = P$$

$$P^2 = A(A^T A)^{-1} A^T A(A^T A)^{-1} A^T$$

$$= A(A^T A)^{-1} A^T = P$$

$$P^2 = P \quad P^k = P \quad P^k = P$$

3) the eigenvalues of any proj.

matrix are all 1 or 0.

$$P\vec{v} = \lambda \vec{v} \quad P^2 \vec{v} = \lambda P \vec{v} \quad P\vec{v} = \lambda(\lambda \vec{v}) = \lambda^2 \vec{v}$$

$$\lambda^2 \vec{v} = \lambda \vec{v} \quad \lambda^2 = \lambda \Rightarrow \lambda^2 - \lambda = 0$$

$$\Rightarrow \lambda(\lambda - 1) = 0 \quad \lambda = 0, \lambda = 1$$

Singular Value Decomposition

$$A = U \Sigma V^T \quad U: \text{square orthonormal } \mathbb{R}^{m \times m} \quad \Sigma: \text{non-square diagonal } I$$

$$V: \text{square orthonormal } \mathbb{R}^{n \times n}$$

$$\text{wide: } \begin{bmatrix} \vec{a}_1^T \\ \vdots \\ \vec{a}_n^T \end{bmatrix} = \begin{bmatrix} \vec{u}_1^T & \vec{u}_2^T & \vec{u}_3^T \\ \vdots & \vdots & \vdots \end{bmatrix} \begin{bmatrix} \sigma_1 & & 0 \\ & \sigma_r & \\ 0 & & 0 \end{bmatrix} \begin{bmatrix} -\vec{v}_1^T \\ \vdots \\ -\vec{v}_r^T \\ \vdots \\ -\vec{v}_n^T \end{bmatrix} \quad \text{COMPACT} \quad \begin{bmatrix} \vec{a}_1^T \\ \vdots \\ \vec{a}_n^T \end{bmatrix} = \begin{bmatrix} \vec{u}_1^T & \vec{u}_2^T & \vec{u}_3^T \\ \vdots & \vdots & \vdots \end{bmatrix} \begin{bmatrix} \sigma_1 & & 0 \\ & \sigma_r & \\ 0 & & 0 \end{bmatrix} \begin{bmatrix} -\vec{v}_1^T \\ \vdots \\ -\vec{v}_r^T \\ \vdots \\ -\vec{v}_n^T \end{bmatrix} \quad A = U \Sigma V^T$$

FULL

$$\text{tall: } \begin{bmatrix} \vec{a}_1^T \\ \vdots \\ \vec{a}_n^T \end{bmatrix} = \begin{bmatrix} \vec{u}_1^T & \vec{u}_2^T & \vec{u}_3^T \\ \vdots & \vdots & \vdots \end{bmatrix} \begin{bmatrix} \sigma_1 & & 0 \\ & \sigma_r & \\ 0 & & 0 \end{bmatrix} \begin{bmatrix} -\vec{v}_1^T \\ \vdots \\ -\vec{v}_r^T \\ \vdots \\ -\vec{v}_n^T \end{bmatrix}$$

OUTER PRODUCT FORM

$$A = \sigma_1 \vec{u}_1 \vec{v}_1^T + \sigma_2 \vec{u}_2 \vec{v}_2^T + \dots + \sigma_r \vec{u}_r \vec{v}_r^T = \sum_{i=1}^r \sigma_i \vec{u}_i \vec{v}_i^T$$

LOW RANK APPROX

$$A = \sigma_1 \vec{u}_1 \vec{v}_1^T + \sigma_2 \vec{u}_2 \vec{v}_2^T + \dots + \sigma_k \vec{u}_k \vec{v}_k^T = \sum_{i=1}^k \sigma_i \vec{u}_i \vec{v}_i^T \quad \sigma_1 > \sigma_2 > \sigma_3$$

Calculation

Tall

$$A^T A \Rightarrow \lambda \text{'s and } \vec{v} \text{'s}$$

$$\sigma_k = \sqrt{\lambda_k}$$

$$\vec{u}_k = \frac{1}{\sigma_k} A^T \vec{v}_k$$

normalize

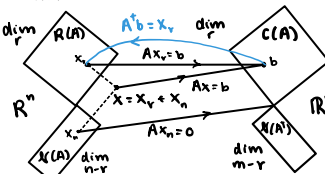
Wide

$$A A^T \Rightarrow \lambda \text{'s and } \vec{u} \text{'s}$$

$$\sigma_k = \sqrt{\lambda_k}$$

$$\vec{v}_k = \frac{1}{\sigma_k} A^T \vec{u}_k$$

Matrix Pseudoinverse



range A: where A actually takes you $\rightarrow \text{col}(A)$

$$A = U \Sigma V^T \quad U^{-1} = U^T \quad b/c \quad U, V \text{ orthogonal matrices}$$

$$A^T = V \Sigma^T U^T \quad V^{-1} = V^T$$

$$\Sigma^+: \text{transpose } \Sigma$$

$$\text{invert along diagonal: } \frac{1}{\sigma_k}$$

Differential & Difference Equations (LTI)

Continuous Time

Scalar Form

$$\ddot{y} - 3\dot{y} + 2y = u(t)$$

$$\dot{x} = ax + bu$$

$$x(t) = e^{at} x(0) + \int_0^t e^{a(t-\tau)} b u(\tau) d\tau$$

State Space VDE Form

$$\dot{\vec{x}}(t) = A \vec{x}(t) + B \vec{u}(t)$$

$$\vec{x}(t) = e^{At} \vec{x}(0) + \int_0^t e^{A(t-\tau)} B \vec{u}(\tau) d\tau$$

Discrete Time

Scalar Form

$$y[i] - 3y[i-1] + 2y[i-2] = u[i]$$

$$x[i+1] = ax[i] + bu[i]$$

$$x[i] = a^i x[0] + \sum_{k=0}^{i-1} a^{i-k-1} bu[k]$$

State Space VDE Form

$$\vec{x}[i+1] = A \vec{x}[i] + B \vec{u}[i]$$

$$\vec{x}[i] = A^i \vec{x}[0] + \sum_{k=0}^{i-1} A^{i-k-1} B \vec{u}[k]$$

$$e^{At} = V e^{\Lambda t} V^{-1} \quad A^i = V \Lambda^i V^{-1}$$

$$e^{At} = \begin{bmatrix} e^{\lambda_1 t} & & \\ & \ddots & \\ & & e^{\lambda_n t} \end{bmatrix} \quad \Lambda^i = \begin{bmatrix} \lambda_1^i & & \\ & \ddots & \\ & & \lambda_n^i \end{bmatrix}$$

$$\underline{x} = V \underline{z} \quad \underline{\dot{z}} = V^{-1} \underline{\dot{x}}$$

$$\underline{z} = V^{-1} \underline{x} \quad \underline{\dot{z}} = V^{-1} \underline{\dot{x}}$$

Don't mix up order!

* Eigenvalues drive long term behavior

Guesses for Particular Solution

$$ae^{bt} \rightarrow Ae^{bt}$$

$$a \cos(bt) + b \sin(bt) \rightarrow A \cos(bt) + B \sin(bt)$$

$$a \cos(bt) \rightarrow A \cos(bt) + B \sin(bt)$$

$$n^{\text{th}} \text{ deg polynomial} \rightarrow A_n t^n + A_{n-1} t^{n-1} + \dots + A_1 t + A_0$$

$$b \sin(bt) \rightarrow A \cos(bt) + B \sin(bt)$$

any triangular matrix: e-values = diagonal values

repeated eigenvalues: find eigenspace $\mathcal{R}(\lambda I - A)$

geometric multiplicity = dim eigenspace

defective matrix NOT diagonalizable

algebraic multiplicity = # repeat values

Spectral Theorem - REAL, SYMMETRIC MATRICES!

$$A = Q \Lambda Q^T = \sum_{i=1}^n \lambda_i \vec{q}_i \vec{q}_i^T \quad Q: \text{orthogonal matrix } Q^T Q = Q Q^T = I$$

Principal Component Analysis * de-mean data first

Data in rows $\Rightarrow \vec{v}$ are PC's

Data in columns $\Rightarrow \vec{u}$ are PC's

$$\vec{x}_L = U_L^T \vec{y} = \begin{bmatrix} \vec{u}_1^T \vec{y} \\ \vdots \\ \vec{u}_L^T \vec{y} \end{bmatrix}$$

$$\vec{x}_{\text{recon}} = U_L \vec{x}_L = U_L U_L^T \vec{x} = \begin{bmatrix} \vec{u}_1 \vec{u}_1^T \vec{x} \\ \vdots \\ \vec{u}_L \vec{u}_L^T \vec{x} \end{bmatrix}$$

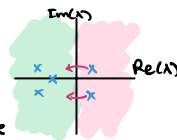
Stability & Feedback Control

Internal Stability

CT: all $\sigma_i < 0 \Rightarrow$ stable

any $\sigma > 0 \Rightarrow$ unstable

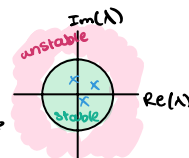
all $\sigma_i \leq 0$ and some $\sigma_i = 0$ } marginally stable or unstable



DT: all $R_i < 1 \Rightarrow$ stable (inward spiral)

any $R > 1 \Rightarrow$ unstable (outward spiral)

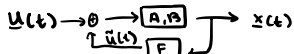
all $R_i \leq 1$ and some $R_i = 1$ } marginally stable or unstable (circle)



BIBO Stability

if for every bounded input and initial state, the output is also bounded. otherwise, system is BIBO unstable.

Feedback Control



Matrix Factorization

$$1. A = QR$$

$$2. A = V\Lambda V^{-1} \text{ square matrices } A, \text{ invertible } V$$

$$3. A = Q\Lambda Q^T = Q\Lambda Q^{-1} \text{ symmetric } A$$

$$4. A = U\Sigma V^T \text{ any matrix!}$$

Converting to State Space

$$\ddot{y} + 3\dot{y} + 2y = 0$$

$$x_1 = y, \quad \dot{x}_1 = x_2$$

$$x_2 = \dot{y}, \quad \dot{x}_2 = \ddot{y} = -3\dot{y} - 2y = -2x_1 - 3x_2$$

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \mathbf{x}$$

Review Session Problems

$$\begin{aligned} \textcircled{1} \begin{vmatrix} \lambda+30 & -10 \\ -10 & \lambda+30 \end{vmatrix} &= (\lambda+30)^2 - 100 = 0 \\ \lambda^2 + 60\lambda + 900 - 100 &= 0 \\ \lambda^2 + 60\lambda + 800 &= 0 \\ (\lambda+40)(\lambda+20) &= 0 \\ \lambda_1 &= -40, \lambda_2 = -20 \end{aligned}$$

A* properties

$$\textcircled{1} \text{ If } A^{-1} \text{ exists: } A^* = A^{-1}$$

$$\textcircled{2} (A^*)^* = A$$

$$\textcircled{3} (A^*)^T = (A^T)^*$$

$$\textcircled{4} (kA)^* = kA^*$$

$$\textcircled{5} AA^*A = A$$

$$\textcircled{6} A^*AA^* = A^*$$

$$\textcircled{7} AA^* = UU_r^T \rightarrow \text{projects onto } C(A)$$

$$\textcircled{8} A^*A = VV_r^T \rightarrow \text{projects onto } C(A^T)$$

$$\text{Overdetermined (Least Squares): } A^* = (A^T A)^{-1} A^T$$

$$\text{Undetermined (Min Norm): } A^* = A^T (A A^T)^{-1}$$

$$\mathbf{x}(t) = V\mathbf{z}(t)$$

$$\mathbf{z}(t) = V^{-1}\mathbf{x}(t)$$

$$\mathbf{z}(0) = V^{-1}\mathbf{x}(0) = \frac{1}{2} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \mathbf{0}$$

$$\mathbf{x}(t) = \begin{bmatrix} -30 & 10 \\ 10 & -30 \end{bmatrix} \mathbf{x}(t) + \begin{bmatrix} 10^3 \\ 0 \end{bmatrix} u(t), \quad \mathbf{x}(t) = V\mathbf{z}(t)$$

$$V\mathbf{z}(t) = \begin{bmatrix} -30 & 10 \\ 10 & -30 \end{bmatrix} V\mathbf{z}(t) + \begin{bmatrix} 10^3 \\ 0 \end{bmatrix} u(t), \quad \mathbf{z}(t) = V^{-1}\mathbf{x}(t)$$

$$\mathbf{z}(t) = V^{-1} \begin{bmatrix} -30 & 10 \\ 10 & -30 \end{bmatrix} V\mathbf{z}(t) + V^{-1} \begin{bmatrix} 10^3 \\ 0 \end{bmatrix} u(t)$$

$$\mathbf{z}(t) = \Lambda \mathbf{z}(t) + \frac{1}{2} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} u(t) = \begin{bmatrix} -40 & 0 \\ 0 & -20 \end{bmatrix} \mathbf{z}(t) + \begin{bmatrix} 500 \\ 0 \end{bmatrix} u(t) = \begin{bmatrix} -40 & 0 \\ 0 & -20 \end{bmatrix} \mathbf{z}(t) + \begin{bmatrix} 500 \\ 0 \end{bmatrix} u(t)$$

$$\mathbf{z}_1(t) = -40\mathbf{z}_1(t) + 5$$

$$\mathbf{z}_{1,u}(t) = C_1 e^{-40t}$$

$$\mathbf{z}_1(t) = C_1 e^{-40t} + \frac{1}{40}$$

$$\mathbf{z}_1(0) = C_1 e^{-40(0)} + \frac{1}{40} = 0$$

$$C_1 = -\frac{1}{40}$$

$$\mathbf{z}_1(t) = -\frac{1}{40} e^{-40t} + \frac{1}{40}$$

$$\textcircled{2} I_0 R_F - V_T \ln(I_0) = V_B - V_T \ln(I_0)$$

$$V_T \ln(I_0) - V_T \ln(I_0) = V_B - I_0 R_F$$

$$V_T \ln\left(\frac{I_0}{I_0}\right) = V_B - I_0 R_F$$

$$\ln\left(\frac{I_0}{I_0}\right) = \frac{V_B - I_0 R_F}{V_T}$$

$$\frac{I_0}{I_0} = e^{\frac{V_B - I_0 R_F}{V_T}} \Rightarrow I_0 = I_0 e^{\frac{V_B - I_0 R_F}{V_T}} \checkmark$$

$$D\hat{p} = \hat{s}$$

$$\hat{p} = (D^T D)^{-1} D^T \hat{s} \Rightarrow \text{yields most optimal values } \hat{p} \text{ and } \hat{s} \text{ for } \ln(I_0) \text{ and } R_F$$

$$\hat{p}_1 = \ln(I_0) \Rightarrow I_0 = e^{\hat{p}_1}$$

$$\hat{p}_2 = R_F \Rightarrow R_F = \hat{p}_2$$

$$\textcircled{3} \|A\tilde{x}\|^2 = (A\tilde{x})^T (A\tilde{x})$$

$$= \tilde{x}^T A^T A \tilde{x}$$

$$= \tilde{x}^T (U\Sigma V^T)^T (U\Sigma V^T) \tilde{x}$$

$$= \tilde{x}^T V \Sigma^T U^T U \Sigma V^T \tilde{x} = \tilde{x}^T V \Sigma^T \Sigma V^T \tilde{x}$$

$$\|V^T \tilde{x}\|^2 = (V^T \tilde{x})^T (V^T \tilde{x})$$

$$= \tilde{x}^T V V^T \tilde{x} \quad \checkmark \text{ is orthonormal } \Rightarrow V V^T = I$$

$$= \tilde{x}^T \tilde{x} = \|\tilde{x}\|^2$$

$$\textcircled{4} \lambda_1 = 20, \lambda_2 = 8$$

$$\sigma_1 = \sqrt{\lambda_1}, \sigma_2 = \sqrt{\lambda_2}$$

$$= \sqrt{20} = 2\sqrt{5}, \quad = \sqrt{8} = 2\sqrt{2}$$

$$\tilde{u}_1 = \begin{bmatrix} 20 & -14 \\ -6 & 20 & -14 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 20 & -14 \\ -6 & 20 & -14 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \tilde{u}_1 \begin{bmatrix} \tilde{u}_1 \\ \tilde{u}_2 \end{bmatrix}$$

$$\tilde{x} = \tilde{u}_1 \tilde{z}_1 + \tilde{u}_2 \tilde{z}_2 = \begin{bmatrix} \tilde{u}_1 \\ \tilde{u}_2 \end{bmatrix} \begin{bmatrix} \tilde{z}_1 \\ \tilde{z}_2 \end{bmatrix}$$

$$= \begin{bmatrix} -2 & -1 & 1 \\ -2 & -1 & 1 \end{bmatrix} \begin{bmatrix} \tilde{z}_1 \\ \tilde{z}_2 \end{bmatrix}$$

$$\tilde{z}_1 = \tilde{u}_1^T \tilde{x} = \begin{bmatrix} \tilde{u}_1 \\ \tilde{u}_2 \end{bmatrix}^T \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\tilde{z}_2 = \tilde{u}_2^T \tilde{x} = \begin{bmatrix} \tilde{u}_2 \\ \tilde{u}_3 \end{bmatrix}^T \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\tilde{z}_3 = \tilde{u}_3^T \tilde{x} = \begin{bmatrix} \tilde{u}_3 \end{bmatrix}^T \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\tilde{z}_4 = \tilde{u}_4^T \tilde{x} = \begin{bmatrix} \tilde{u}_4 \end{bmatrix}^T \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\tilde{z}_5 = \tilde{u}_5^T \tilde{x} = \begin{bmatrix} \tilde{u}_5 \end{bmatrix}^T \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\tilde{z}_6 = \tilde{u}_6^T \tilde{x} = \begin{bmatrix} \tilde{u}_6 \end{bmatrix}^T \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\tilde{z}_7 = \tilde{u}_7^T \tilde{x} = \begin{bmatrix} \tilde{u}_7 \end{bmatrix}^T \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\tilde{z}_8 = \tilde{u}_8^T \tilde{x} = \begin{bmatrix} \tilde{u}_8 \end{bmatrix}^T \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\tilde{z}_9 = \tilde{u}_9^T \tilde{x} = \begin{bmatrix} \tilde{u}_9 \end{bmatrix}^T \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\tilde{z}_{10} = \tilde{u}_{10}^T \tilde{x} = \begin{bmatrix} \tilde{u}_{10} \end{bmatrix}^T \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\tilde{z}_{11} = \tilde{u}_{11}^T \tilde{x} = \begin{bmatrix} \tilde{u}_{11} \end{bmatrix}^T \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\tilde{z}_{12} = \tilde{u}_{12}^T \tilde{x} = \begin{bmatrix} \tilde{u}_{12} \end{bmatrix}^T \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\tilde{z}_{13} = \tilde{u}_{13}^T \tilde{x} = \begin{bmatrix} \tilde{u}_{13} \end{bmatrix}^T \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\textcircled{5} \mathbf{x}[i+1] = A\mathbf{x}[i] + B\mathbf{u}[i]$$

$$= A\mathbf{x}[i] + B\mathbf{F}\mathbf{u}[i] = A\mathbf{x}[i] + B\mathbf{B}^{-1}A\mathbf{x}[i]$$

$$= (A+B\mathbf{B}^{-1}A)\mathbf{x}[i]$$

$$A+B\mathbf{B}^{-1}A = \begin{bmatrix} 1.3 & 0.5 \\ 0 & 1.4 \end{bmatrix}$$

$$\lambda_1 = 1.3, \lambda_2 = 1.4, \lambda_1, \lambda_2 > 1 \Rightarrow \text{not BIBO stable}$$

$$\mathbf{x}[i+1] = A\mathbf{x}[i] + B\mathbf{u}[i]$$

$$= A\mathbf{x}[i] + B\mathbf{F}\mathbf{x}[i] = A\mathbf{x}[i] + B\mathbf{B}^{-1}A\mathbf{x}[i]$$

$$= (A+B\mathbf{B}^{-1}A)\mathbf{x}[i] = 2A\mathbf{x}[i]$$

$$2A = \begin{bmatrix} 0.6 & 0.5 \\ 0 & 0.8 \end{bmatrix}, \lambda_1 = 0.6, \lambda_2 = 0.8$$

$$\Rightarrow \lambda_1, \lambda_2 < 1 \text{ BIBO stable}$$